

A, T, G, GPT

Teaching Machines the Language of Life

Aayan Alwani | Grade __ | _____ High School

In 2003, it took **13 years** and **\$2.7 billion** to read one human genome. Today, AI models trained on **9.3 trillion** DNA letters from every domain of life are learning to read genomes the way ChatGPT reads English—and they're beginning to write new ones.

SECTION I

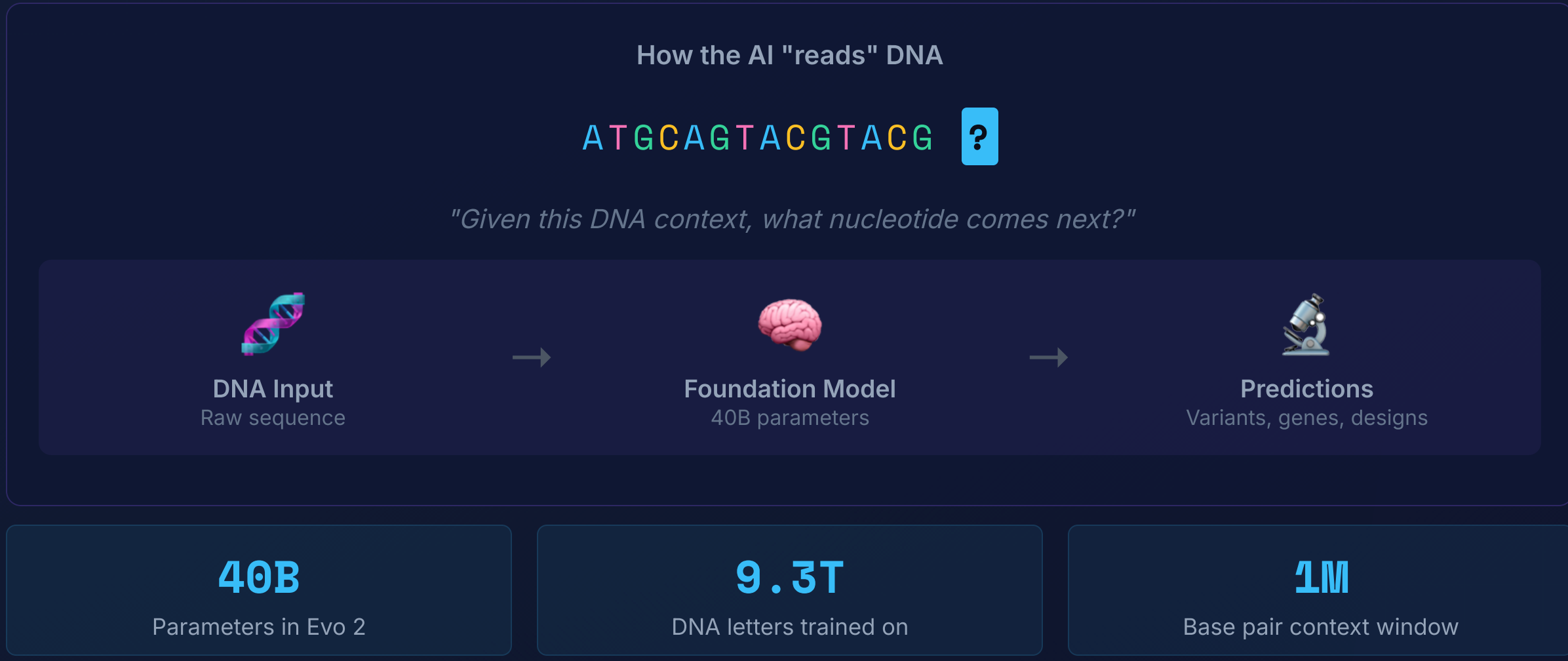
Topic Background

Genomic foundation models are large AI systems trained on billions of DNA sequences (A, T, G, C) from thousands of species. Just as ChatGPT learned the rules of English by reading the internet, these models learn the **"grammar" of genomes**—which patterns regulate genes, how mutations affect function, and what makes DNA from a human different from a bacterium.

The core idea is **next-token prediction**: given a stretch of DNA, predict what nucleotide comes next. By training on trillions of bases, the AI builds a deep internal model of how genomes work—without being explicitly taught biology.

These models can then **predict whether a mutation causes disease**, design synthetic genes with specific functions, annotate newly sequenced genomes, and transfer knowledge across species—all without running a single wet-lab experiment. Every prediction still requires experimental validation, but the speed of discovery is transformed.

ChatGPT	Evo 2
Trained on text from the internet	Trained on DNA from all life on Earth
Learns grammar & meaning	Learns genetic "grammar"
Predicts the next word	Predicts the next nucleotide
Generates new text	Generates new DNA sequences



SECTION II

Technology Timeline



SECTION III

Biotech Innovators & Economic Impact

Evo & Evo 2 Arc Institute • Stanford • UC Berkeley Open-source models led by Patrick Hsu and Brian Hie . Evo 2 predicts variant pathogenicity across the human genome. Arc Institute was co-founded with a \$650M endowment by Stripe CEO Patrick Collison. Free & Open Source	Nucleotide Transformer BioNTech / InstaDeep Deployed in mRNA vaccine and cancer therapeutic design. BioNTech acquired InstaDeep for ~\$680M in 2023—one of the largest AI-biotech acquisitions in history. \$680M Acquisition	BioNeMo Platform NVIDIA GPU infrastructure powering genomic AI across pharma. Partners include Genentech/Roche and Amgen. The AI-in-genomics market is projected to reach \$5-10B by 2030 . \$5-10B Market by 2030	Illumina The Data Engine Sequencing machines that generate virtually all the DNA data these models train on. Dropped genome cost from \$2.7B to under \$200. Now investing in AI variant interpretation (SpliceAI, PrimateAI). \$4.5B Revenue ~80% Market Share
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SECTION IV

Ethical, Legal & Social Issues

Biosecurity AI that generates realistic DNA could theoretically design pathogens. The 2024 U.S. Executive Order on AI specifically flagged risks from generative biology. Arc Institute uses responsible-release policies and red-teaming before publication.	Genomic Bias Training data is heavily skewed toward European-ancestry genomes. Models may produce worse predictions for underrepresented populations, risking amplification of existing health disparities in clinical genomics.
Privacy & Consent Biobank samples were collected before anyone imagined AI training. Genomic data can re-identify individuals. The Genetic Information Nondiscrimination Act (GINA) has significant gaps—it does not cover life insurance.	IP & Ownership If AI designs a novel gene sequence, who holds the patent? Current IP law does not address AI-generated biological material. Regulatory frameworks remain untested.
Access & Equity Training a 40B-parameter model requires thousands of GPU-hours—resources only wealthy institutions can afford. Open-sourcing helps, but infrastructure and expertise gaps risk creating a two-tier system in genomic medicine.	

SECTION V

References

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Generative AI (Claude, Anthropic) was used to assist with research and poster content development. All facts were independently verified.

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